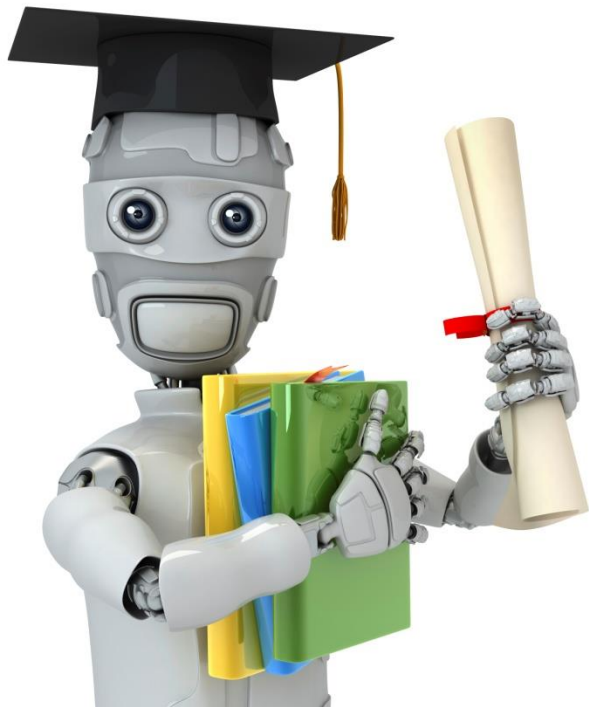


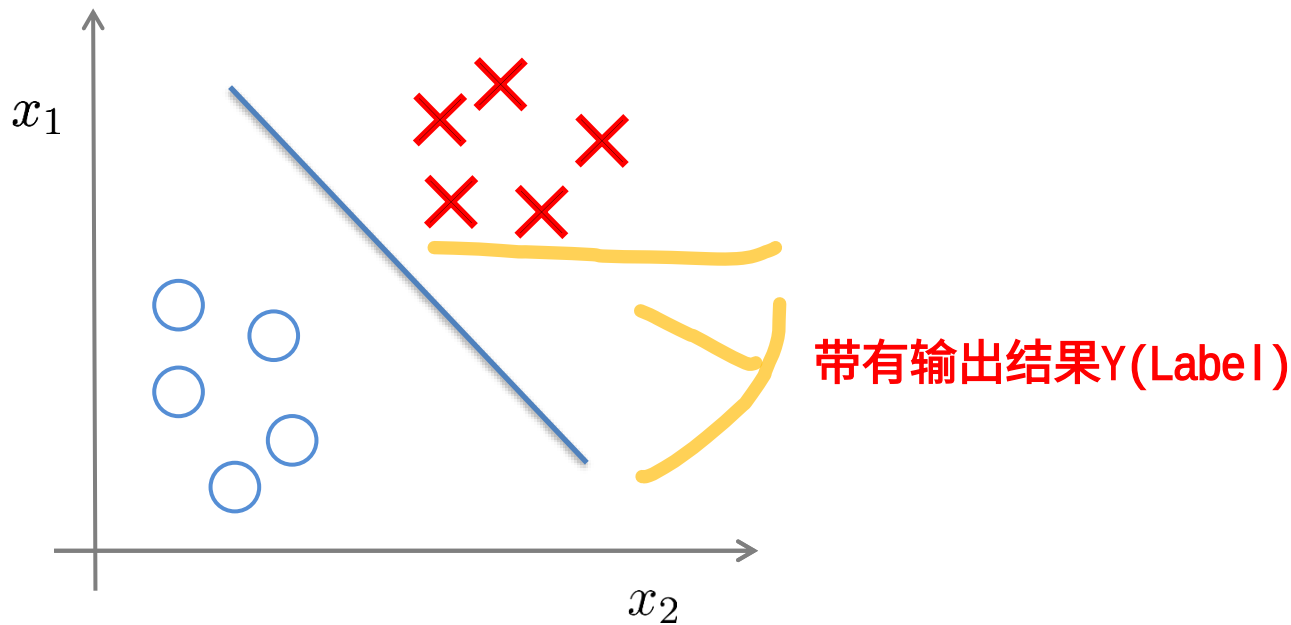


机器学习

聚类

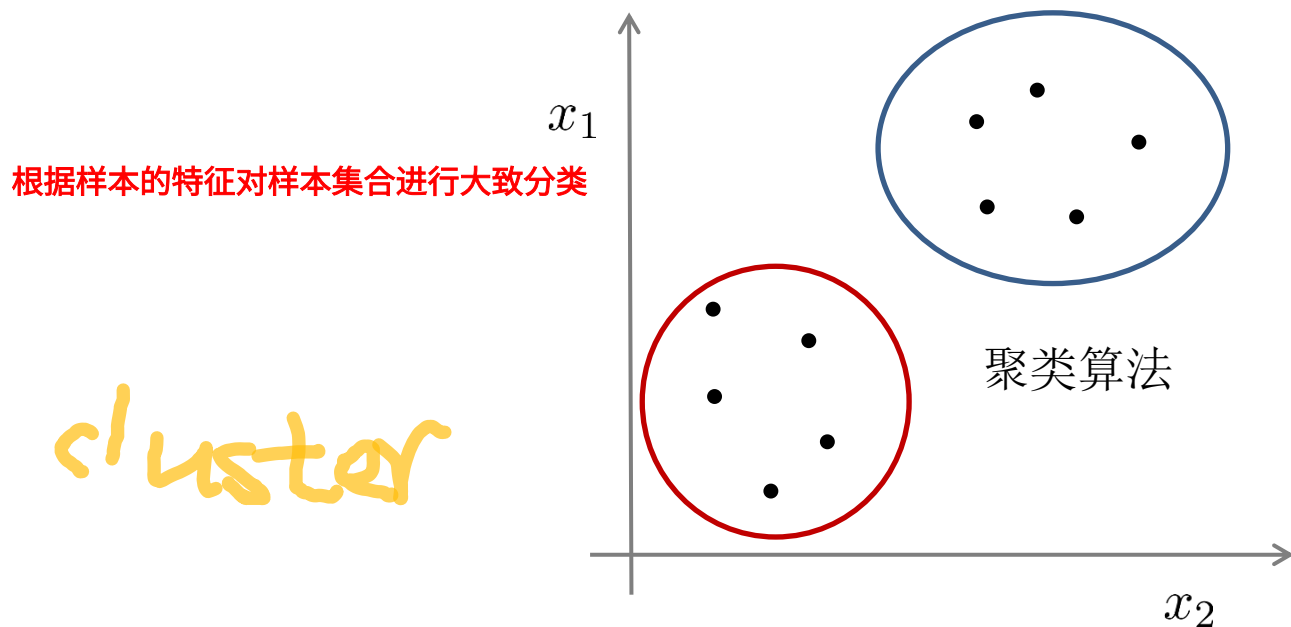
无监督学习简介

监督学习



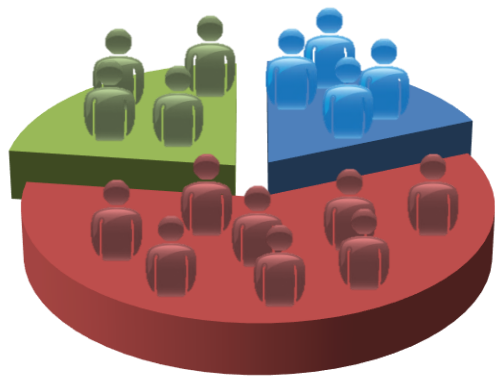
训练集: $\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), (x^{(3)}, y^{(3)}), \dots, (x^{(m)}, y^{(m)})\}$

无监督学习

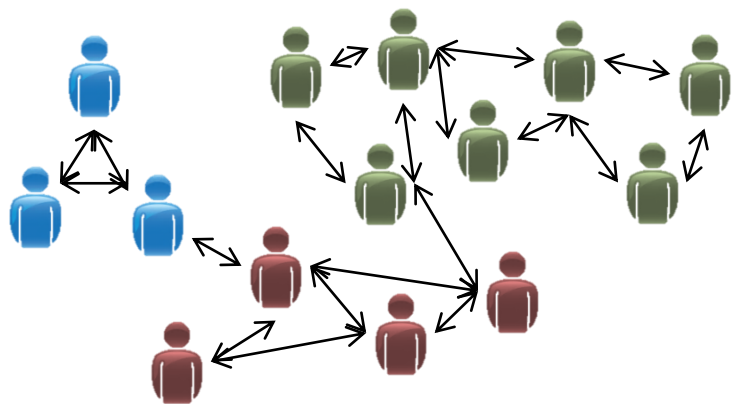


训练集: $\{x^{(1)}, x^{(2)}, x^{(3)}, \dots, x^{(m)}\}$

聚类的应用



市场分割



社交网络分析



组织计算机集群

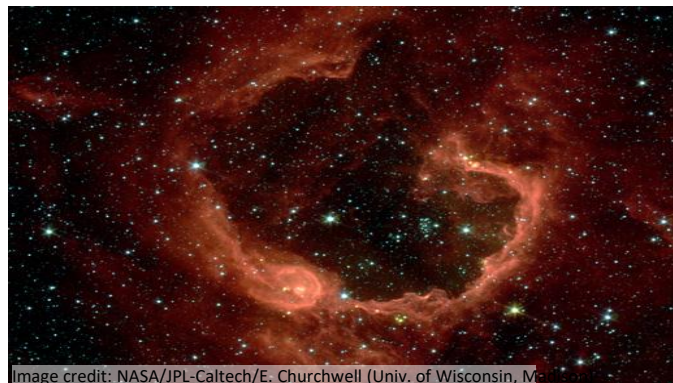
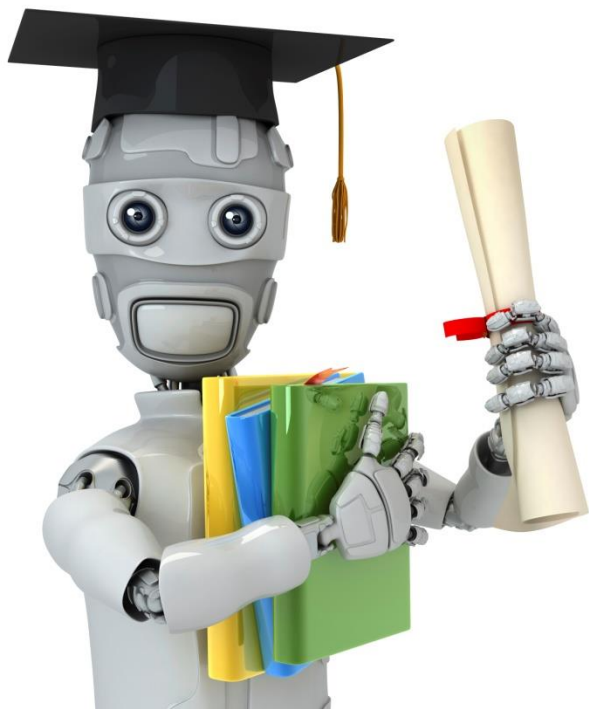


Image credit: NASA/JPL-Caltech/E. Churchwell (Univ. of Wisconsin, M

天文数据分析

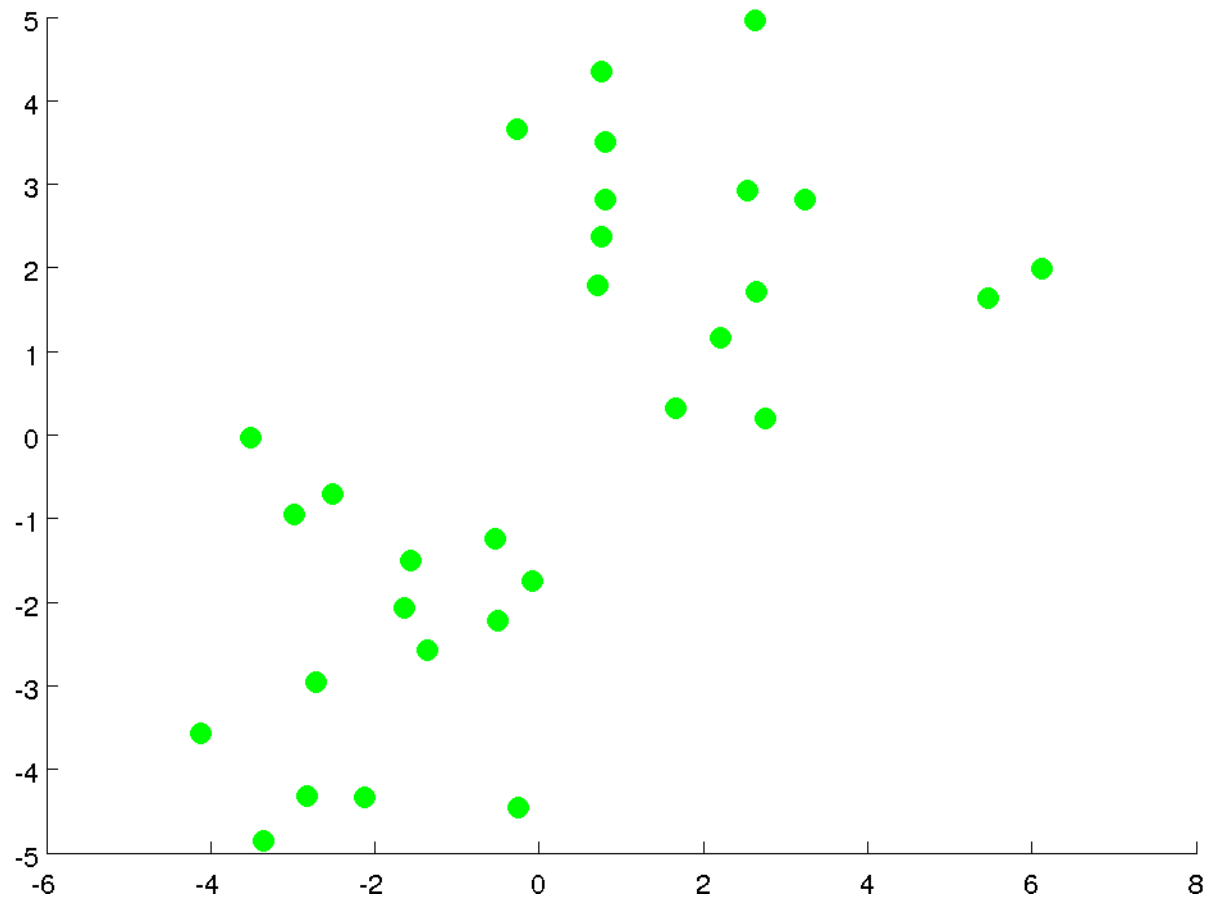


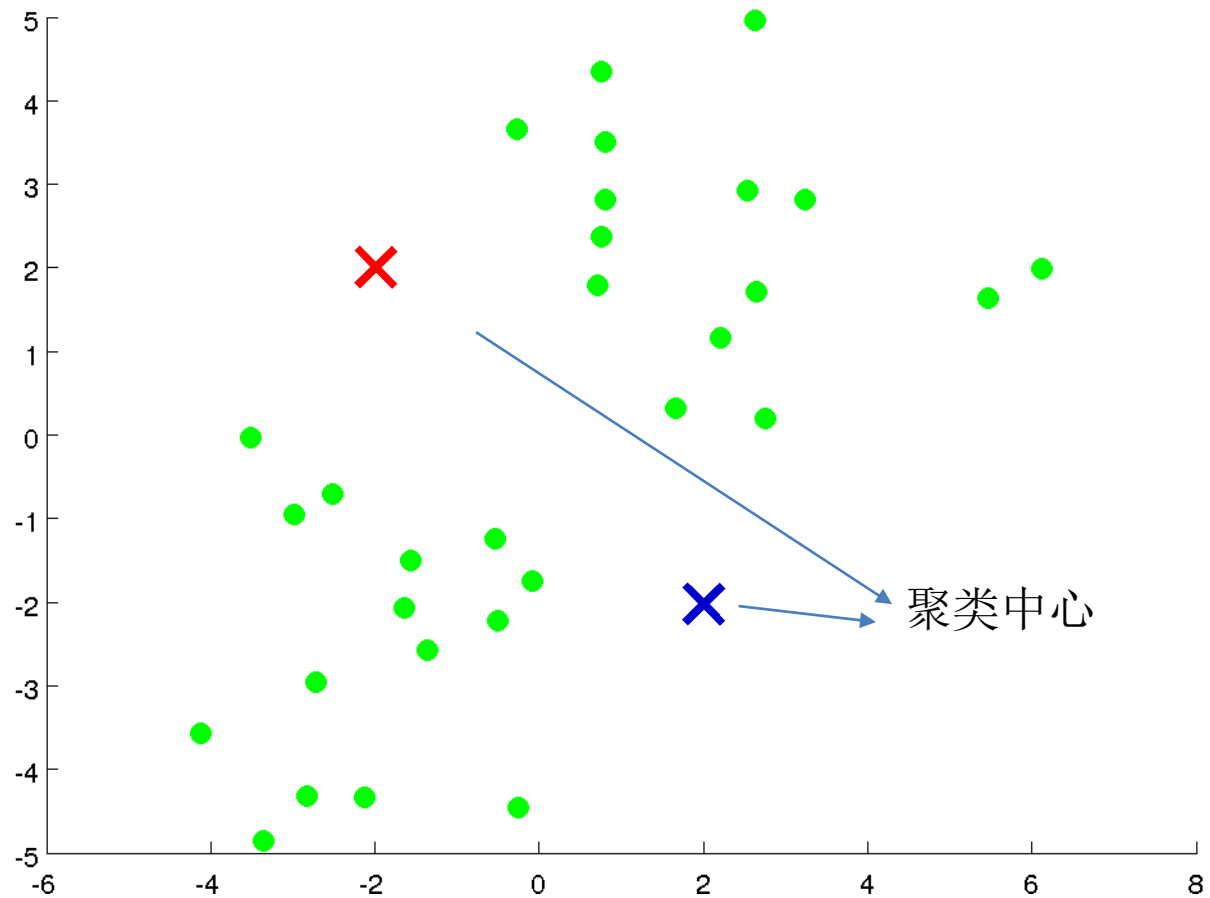
机器学习

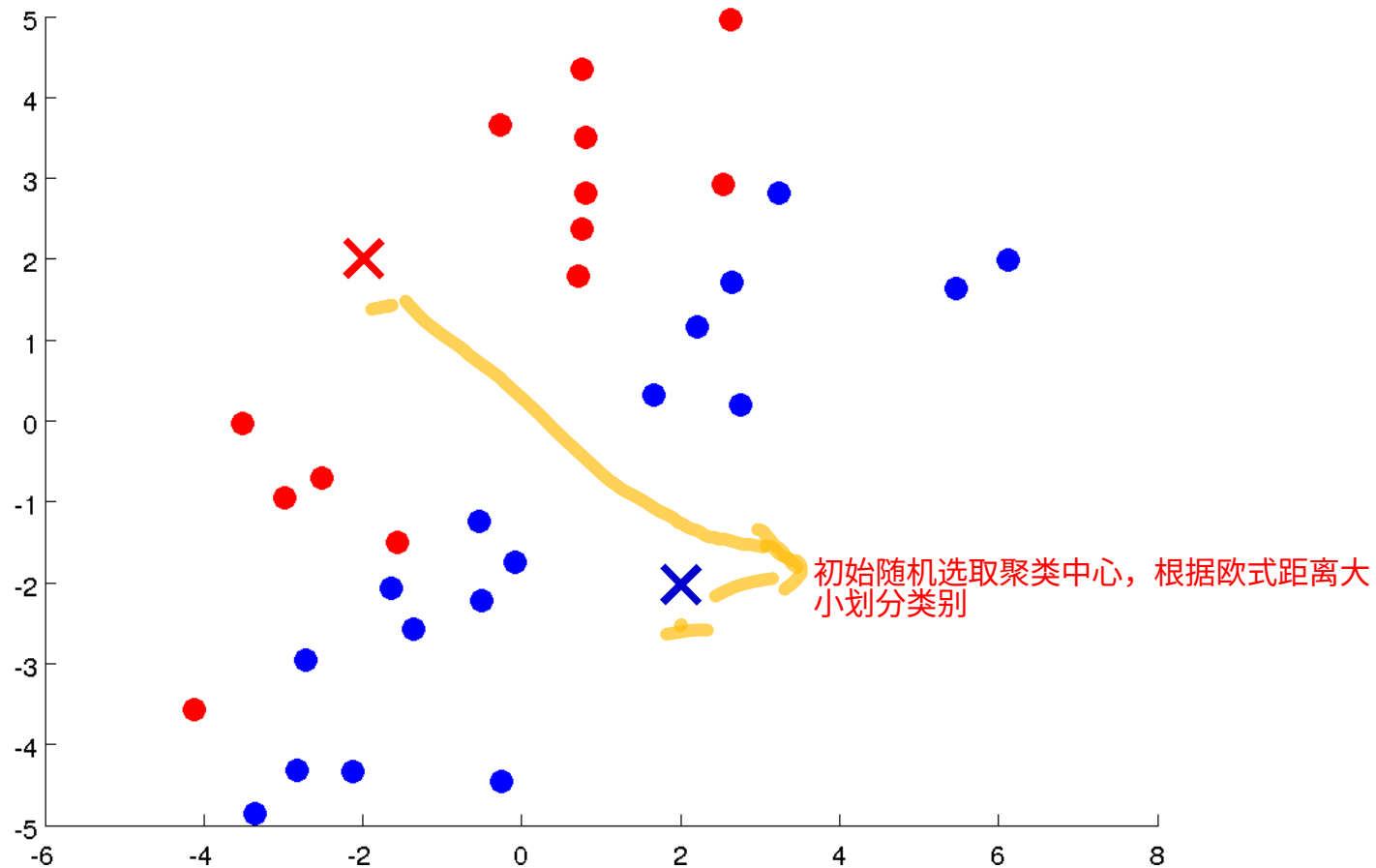
聚类

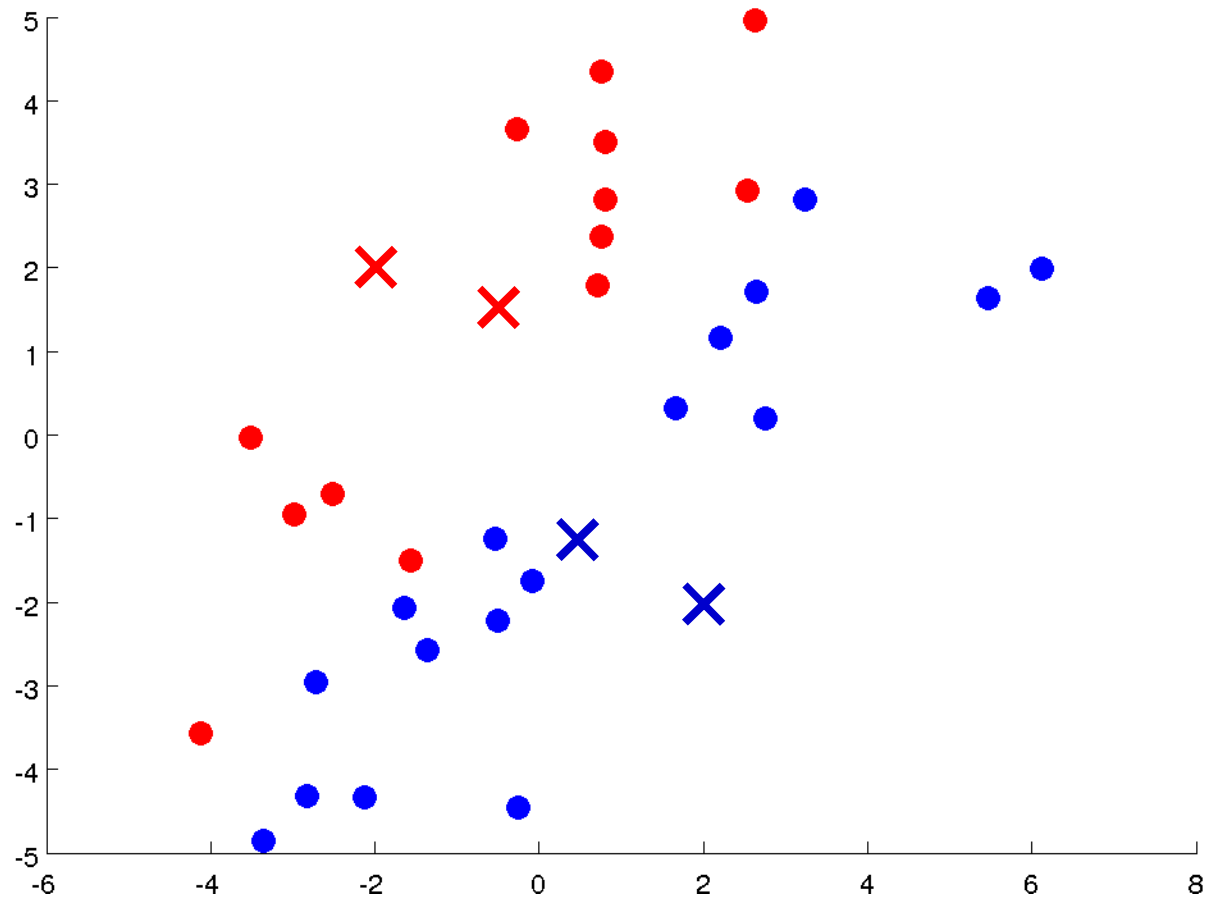
K-均值算法

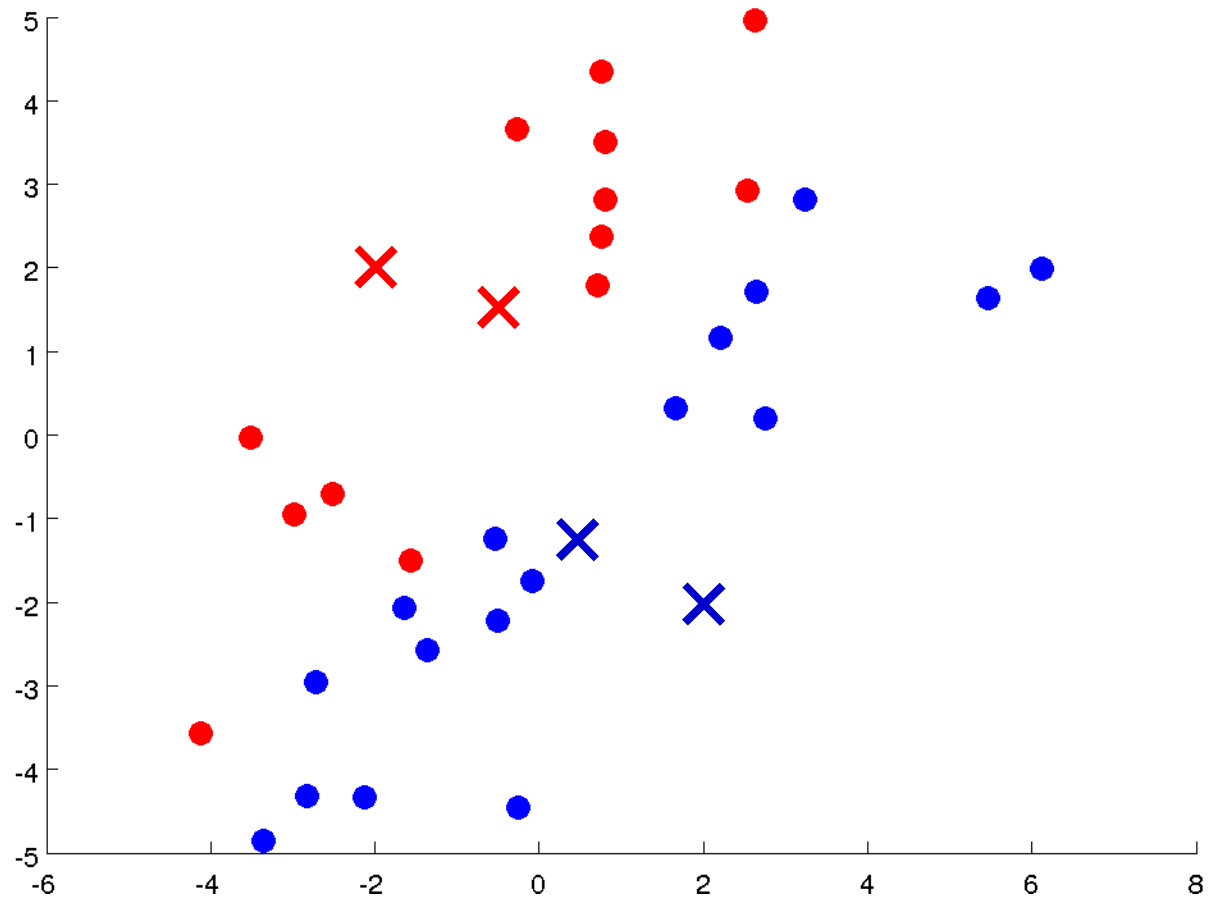
K-means

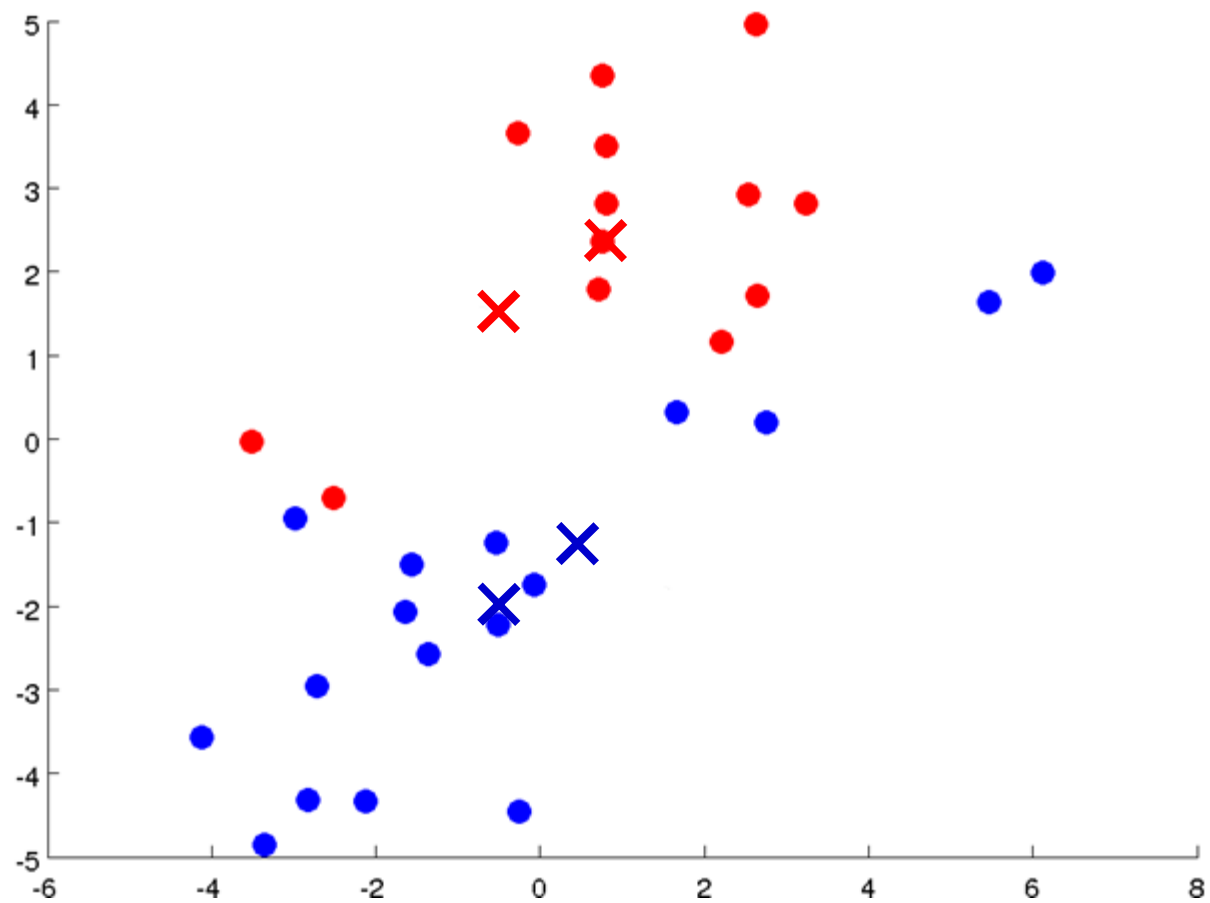


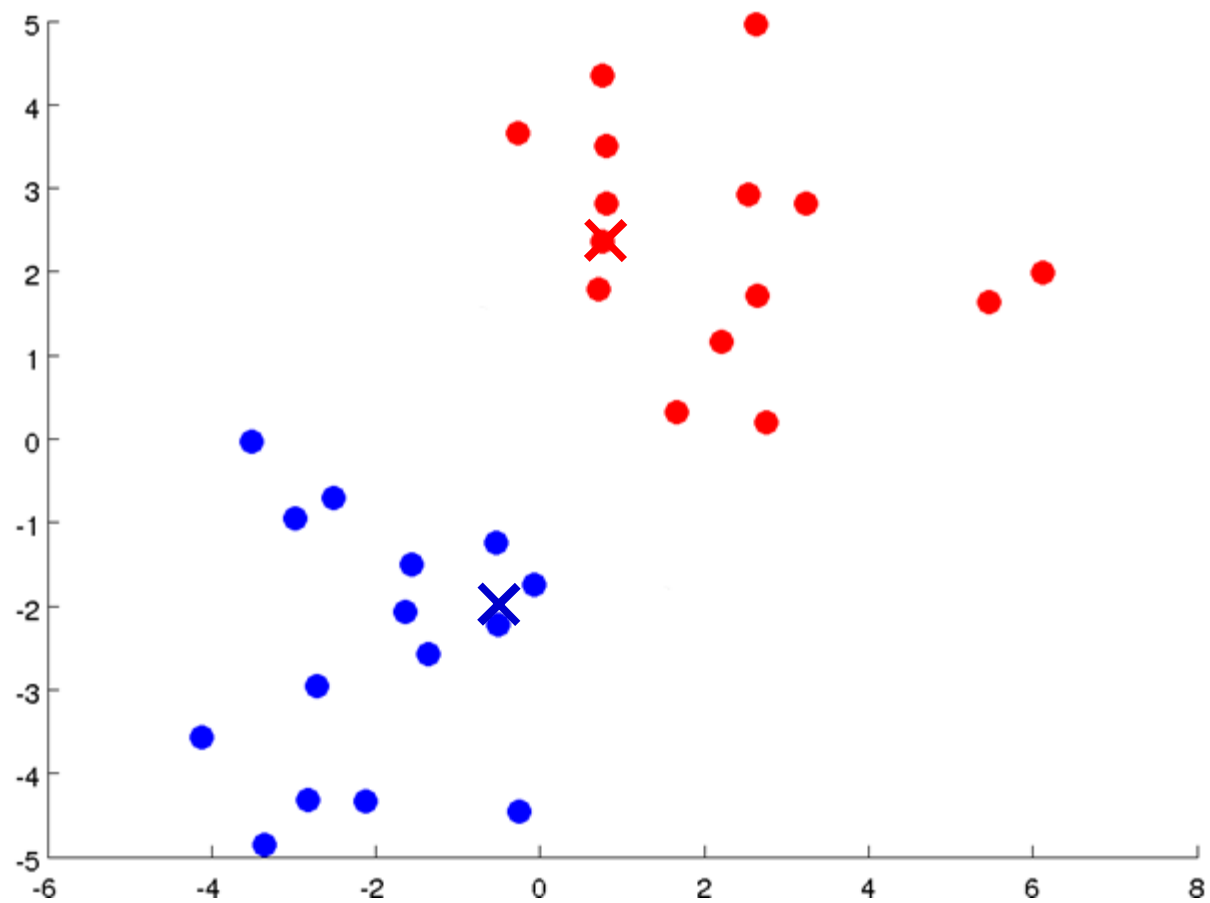


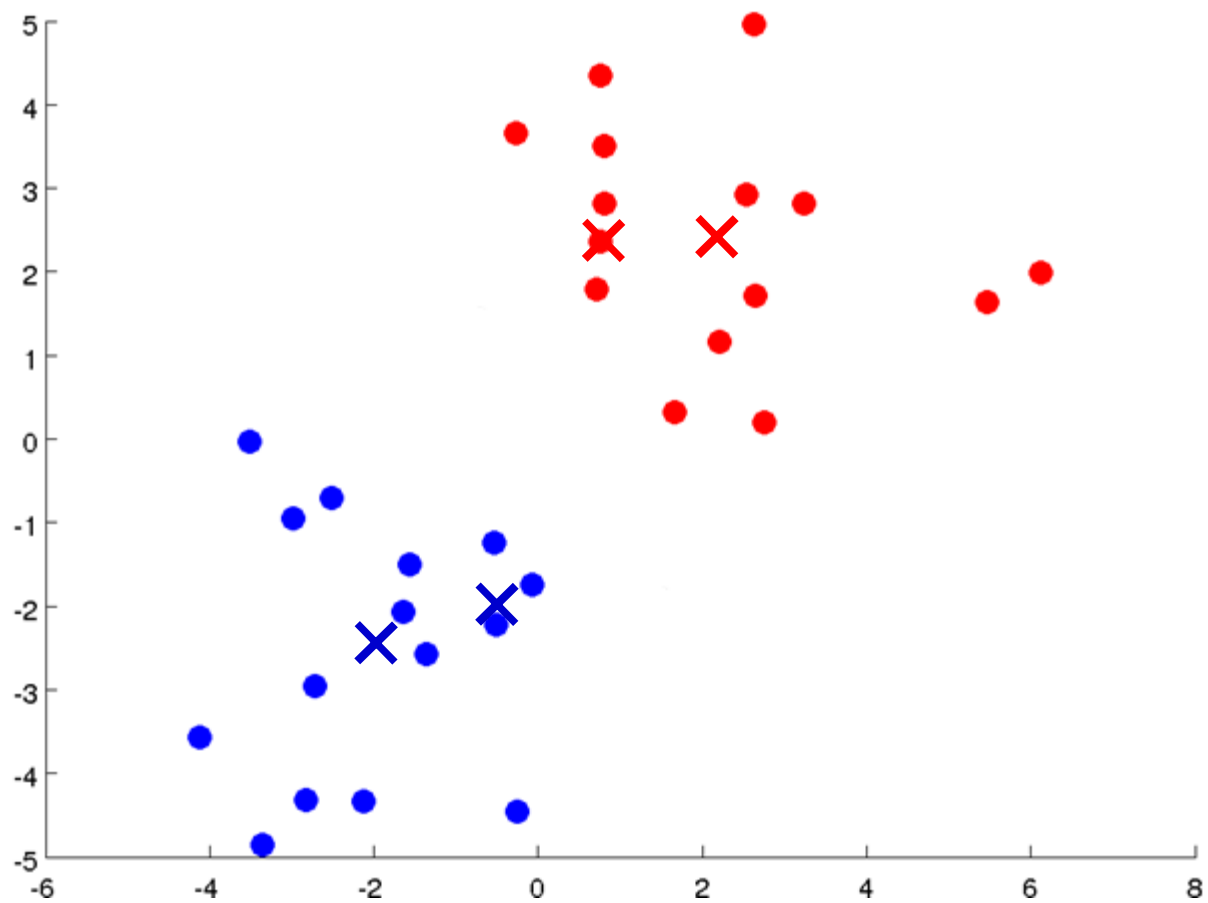


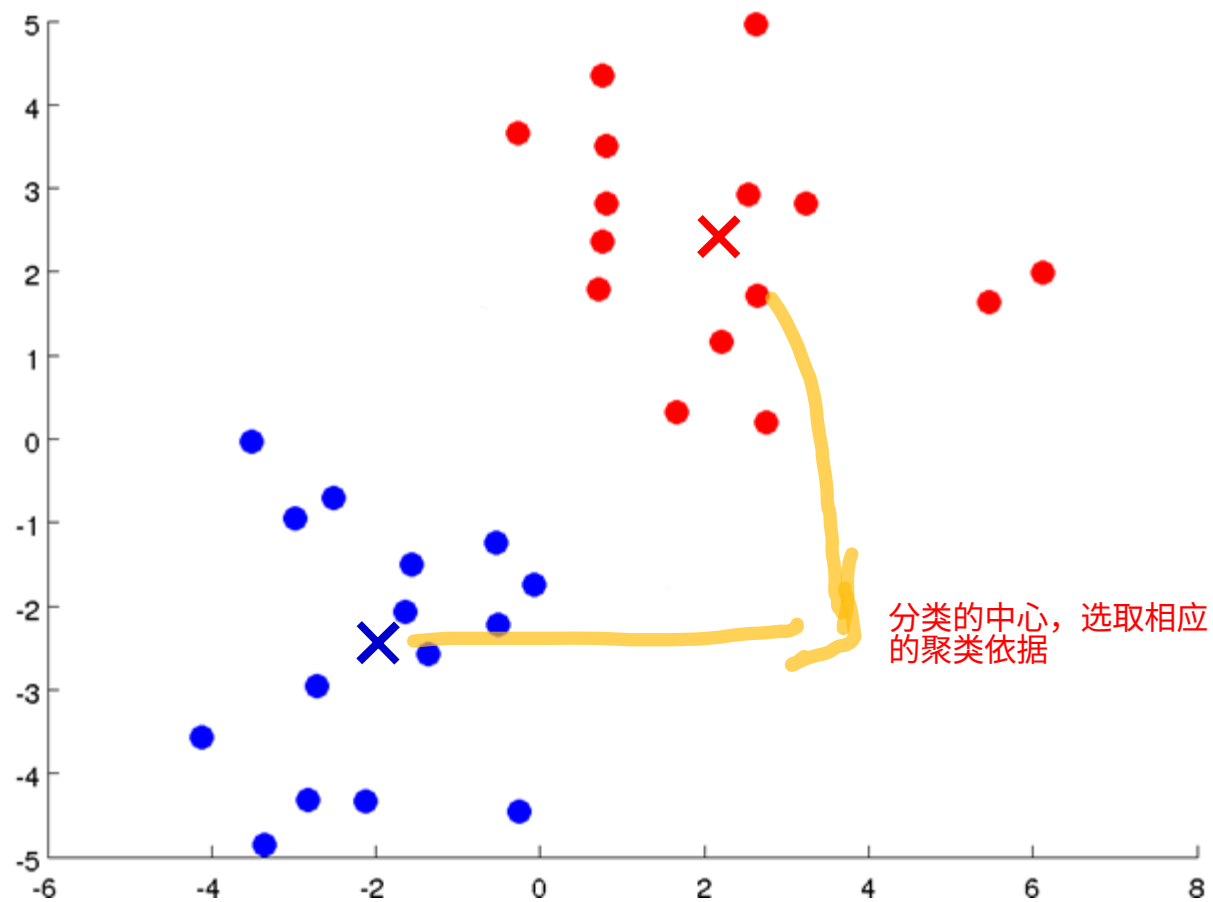












分类的中心，选取相应的聚类依据

K-均值算法

输入:

- K (聚类数)
- 训练集 $\{x^{(1)}, x^{(2)}, \dots, x^{(m)}\}$

每个样本点作为一个向量

$x^{(i)} \in \mathbb{R}^n$ (去掉 $x_0 = 1$)

去掉偏置 θ_0

$m \times n$

K-均值算法

随机初始化 K 个聚类中心点 $\mu_1, \mu_2, \dots, \mu_K \in \mathbb{R}^n$

Repeat {

for $i = 1$ to m

样本集的点按照欧式距离远近归类到对应的类簇cluster

$c^{(i)} :=$ index (from 1 to K) of cluster centroid
closest to $x^{(i)}$

for $k = 1$ to K

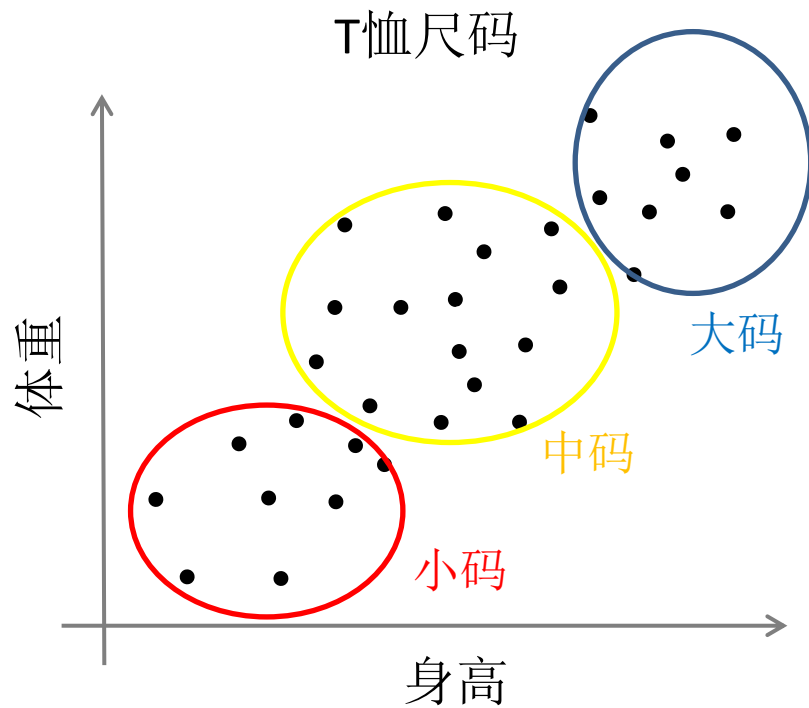
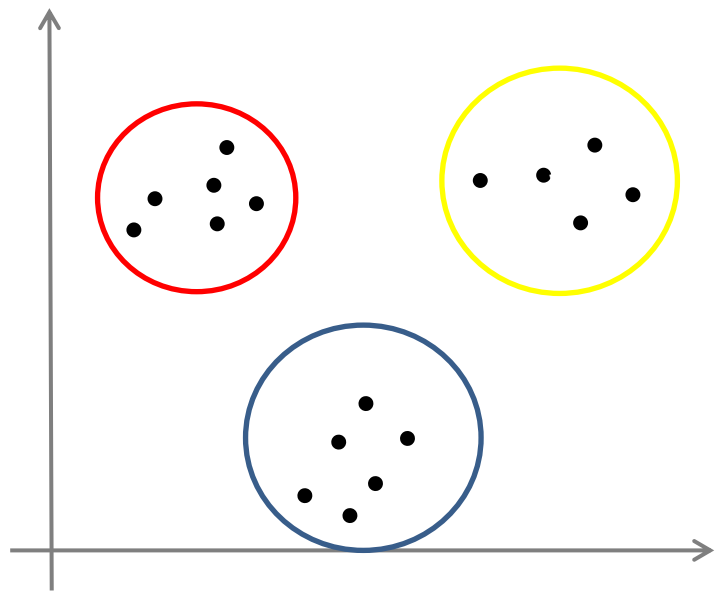
更新每一类的中心点

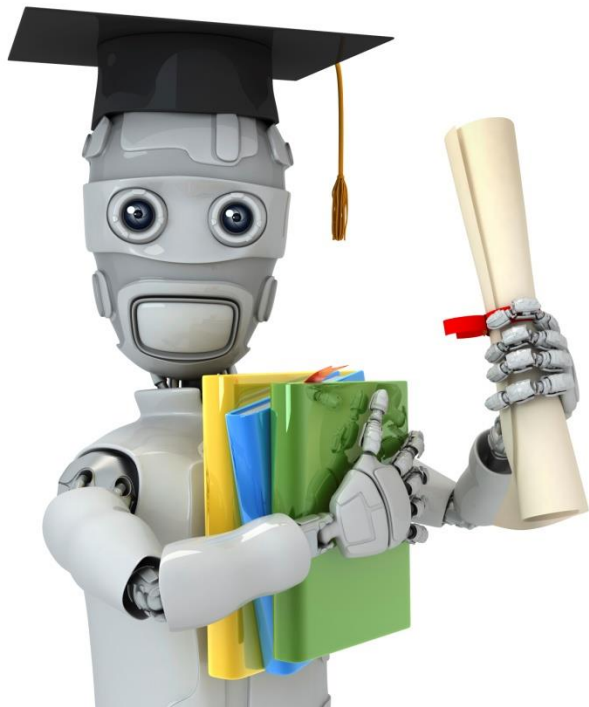
$\mu_k :=$ average (mean) of points assigned to cluster k

}

收敛条件：各个类簇的中心点坐标不再发生变化（给定一个阈值），所有样本点的类簇划分不发生变化

用于非分离聚类的 K-均值算法





机器学习

聚类

优化目标

K-均值优化目标

$c^{(i)}$ = 示例 $x^{(i)}$ 所在的集群 $K(1, 2, \dots)$ 的索引

μ_k = 聚类中心 k ($\mu_k \in \mathbb{R}^n$)

$\mu_{c^{(i)}}$ = 示例 $x^{(i)}$ 所在集群的聚类中心

优化目标:

使得每个点对聚类的中心点的欧式距离最小

$$J(c^{(1)}, \dots, c^{(m)}, \mu_1, \dots, \mu_K) = \frac{1}{m} \sum_{i=1}^m \|x^{(i)} - \mu_{c^{(i)}}\|^2$$

distortion function(失真函数)

类似于类内方差最小

$$\min_{\substack{c^{(1)}, \dots, c^{(m)}, \\ \mu_1, \dots, \mu_K}} J(c^{(1)}, \dots, c^{(m)}, \mu_1, \dots, \mu_K)$$

K-均值算法

随机初始化 K 个聚类中心点 $\mu_1, \mu_2, \dots, \mu_K \in \mathbb{R}^n$

Repeat {

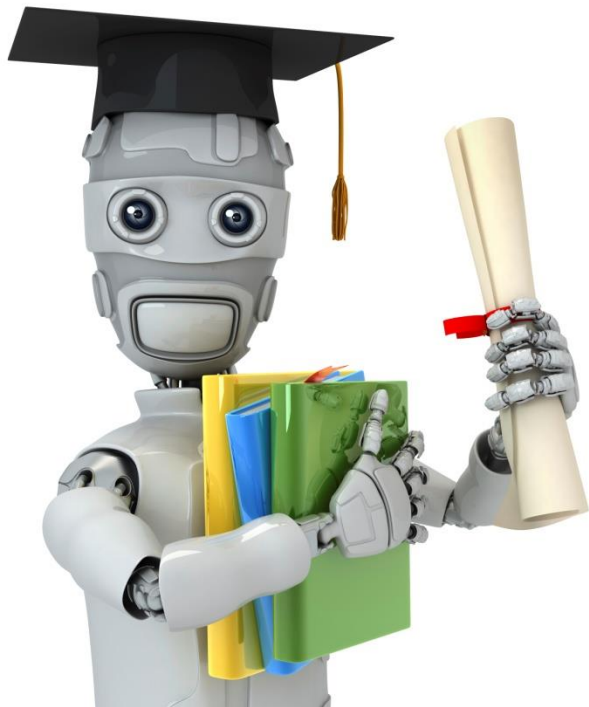
 for $i = 1$ to m

$c^{(i)} :=$ index (from 1 to K) of cluster centroid
 closest to $x^{(i)}$

 for $k = 1$ to K

$\mu_k :=$ average (mean) of points assigned to cluster k

}



机器学习

聚类

随机初始化

K-均值算法

随机初始化 K 个聚类中心点 $\mu_1, \mu_2, \dots, \mu_K \in \mathbb{R}^n$

Repeat {

 for $i = 1$ to m

$c^{(i)} :=$ index (from 1 to K) of cluster centroid
 closest to $x^{(i)}$

 for $k = 1$ to K

$\mu_k :=$ average (mean) of points assigned to cluster k

}

随机初始化

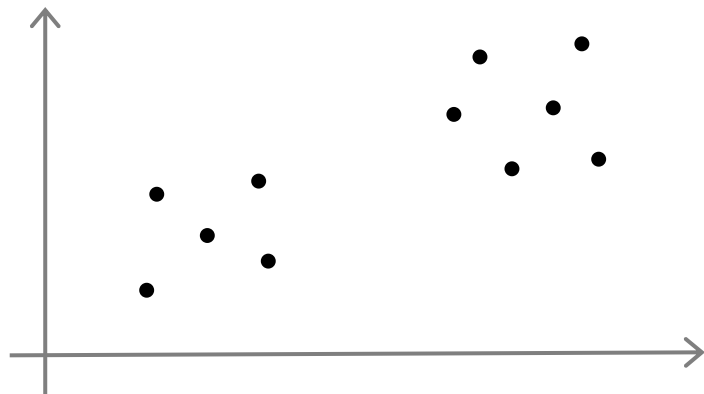
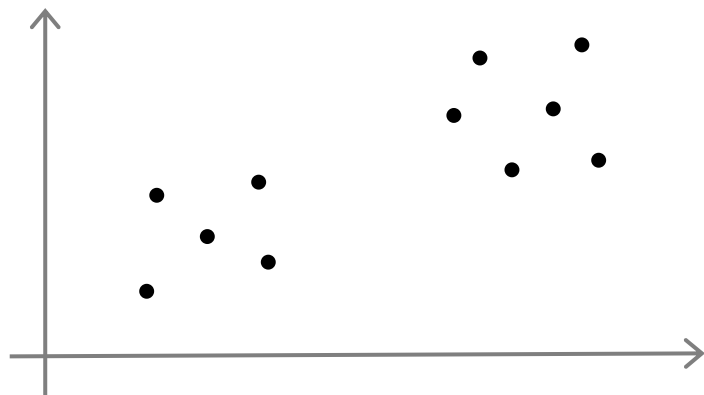
可能导致局部最优，聚类效果不佳

应该选择 $K < m$

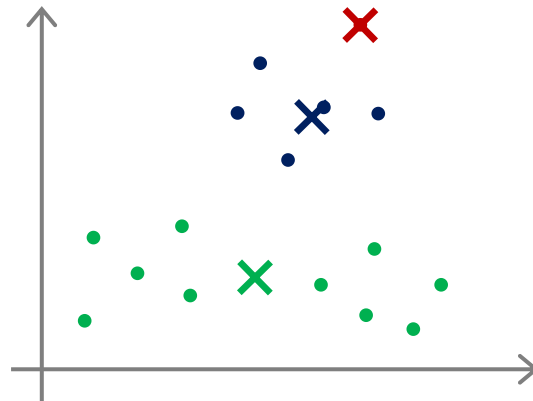
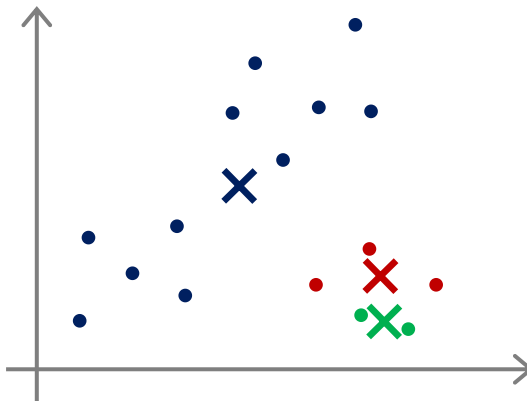
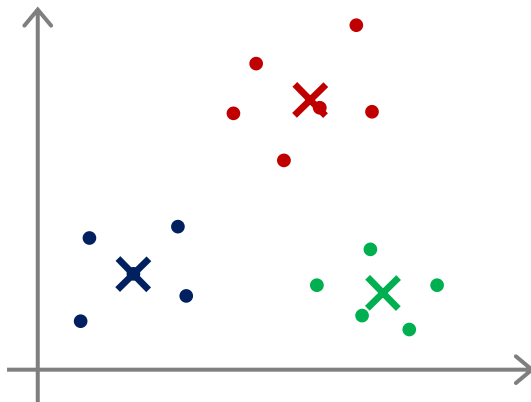
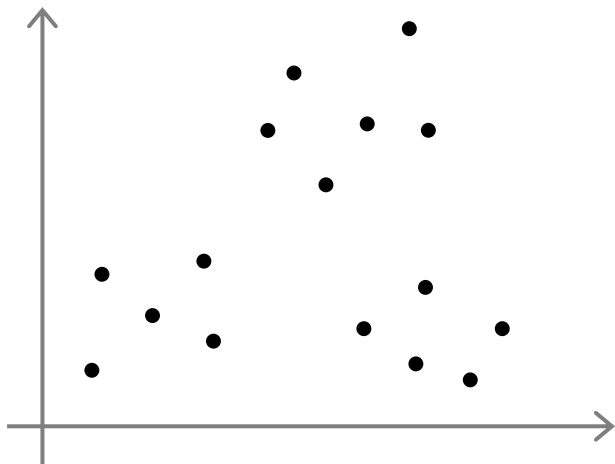
样本点

随机选择 K 个训练实例

设置 $\mu_1, \dots, \mu_K$ 等于这 K 个样本



局部最优



随机初始化

迭代次数

For $i = 1$ to 100 {

随机初始化K-均值.

或者随机取k个点作为聚类中心

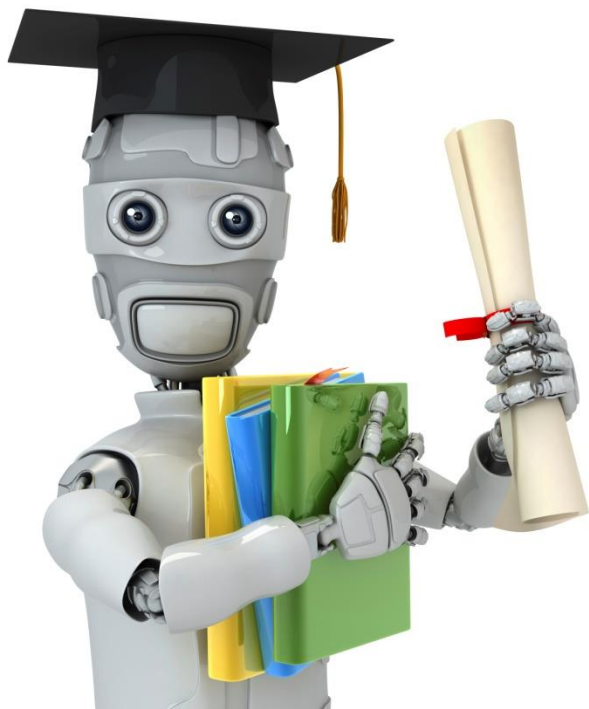
运行K-均值. 得到 $c^{(1)}, \dots, c^{(m)}, \mu_1, \dots, \mu_K$.

计算代价函数 (失真)

$$J(c^{(1)}, \dots, c^{(m)}, \mu_1, \dots, \mu_K)$$

}

选择在代价 $J(c^{(1)}, \dots, c^{(m)}, \mu_1, \dots, \mu_K)$ 中最低的聚类



机器学习

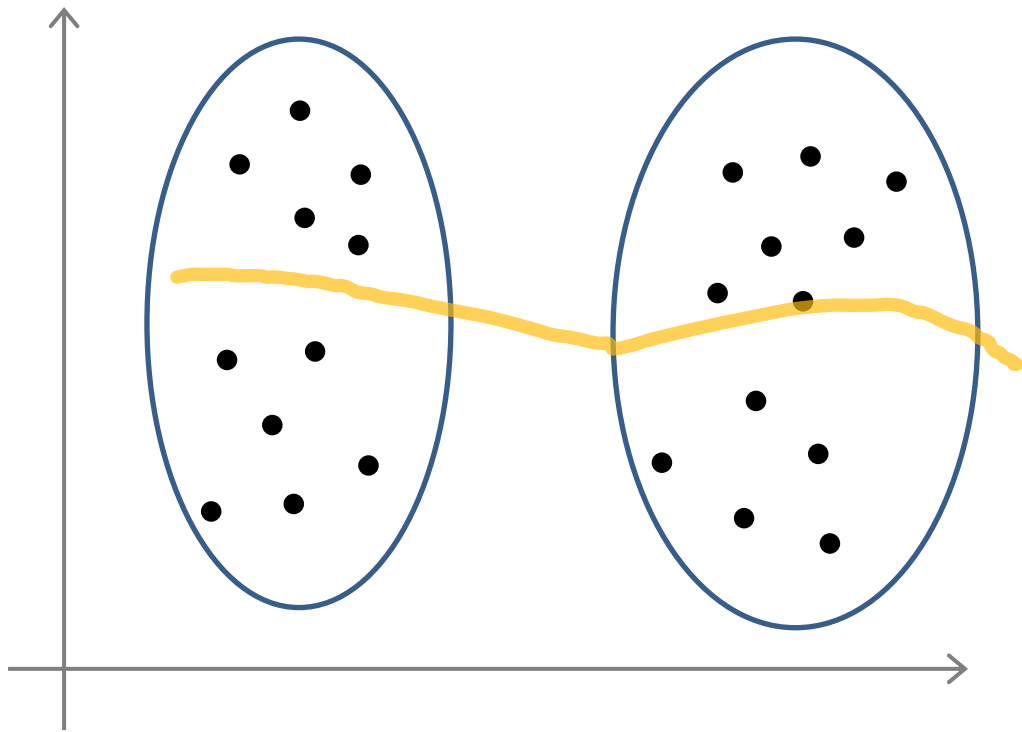
聚类

选择聚类数



超参数hyper parameter (固定值)

K的正确值是多少？

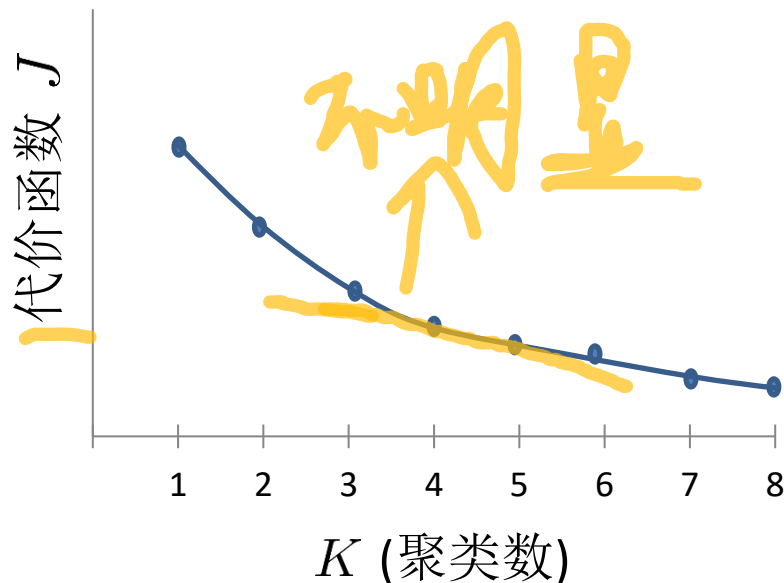
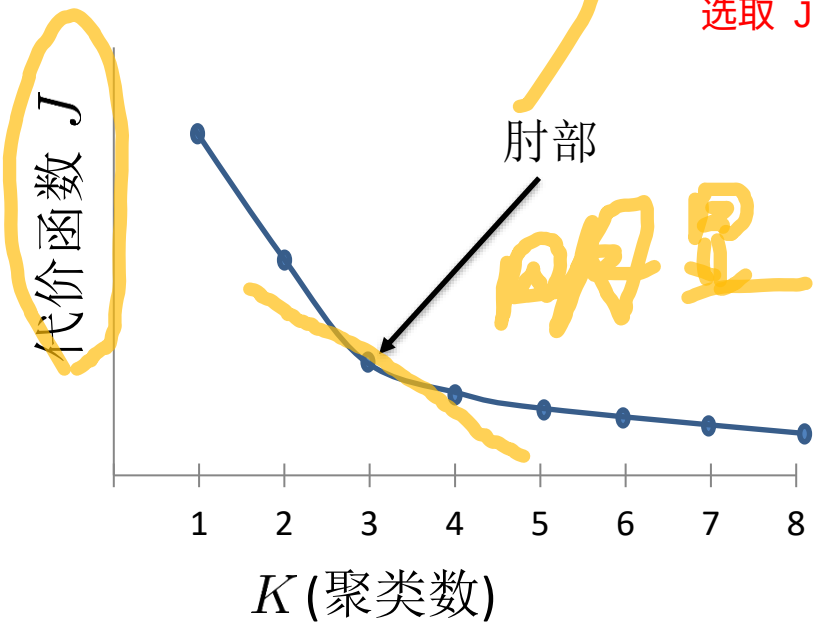


选择K的值

肘部法则:

手肘法

聚类数目越多, 平均距离 J 越小;
选取 J 下降速率突变的点作为分类数目



选择K的值

有时候，您会运行K-均值算法来获取聚类，以便用于某些后续或下游目的。在评估K-均值时，应该基于它在后续目的中的性能表现来进行评估，使用相应的评估指标。

例如：

